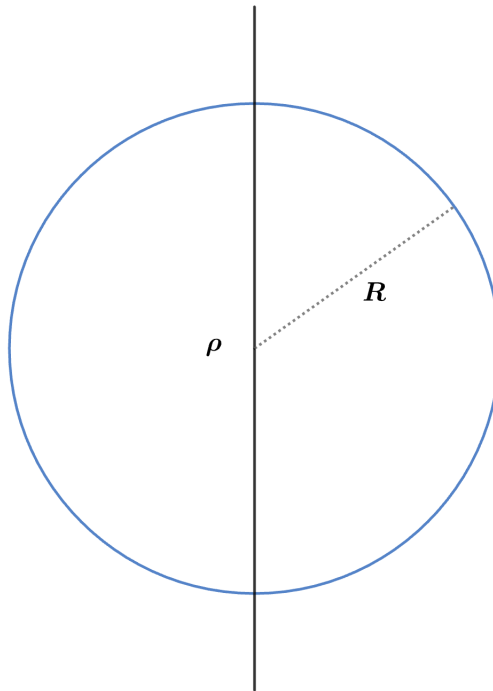


2007 F=ma Exam: Problem 21

Kevin S. Huang



Recall the moment of inertia of a sphere is

$$I = \frac{2}{5}MR^2 \propto MR^2$$

Since $M \propto \rho R^3$,

$$I \propto \rho R^5$$

If the density ρ is the same and the radius R is doubled, the moment of inertia I increases by a factor of $2^5 = 32$ so the answer is E.